

Total No. of Questions : 8]

SEAT No. :

PA-2674

[Total No. of Pages : 2

[5927] 463

B.E.

**HONOURS IN ARTIFICIAL INTELLIGENCE & MACHINE
LEARNING
(2019 Pattern) (Semester - VII) (410301)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is out of bag samples in random forest? Explain it with example. [8]
b) Explain with example the variant of SVM, the Support vector regression. [9]

OR

- Q2)** a) State and explain different types of kernel functions in SVM? [9]
b) What is random Forest? Write an algorithm for random forest. [8]

- Q3)** a) Explain Bayesian View of Learning and Dimensionality Reduction neural network. [9]
b) What is perceptron? Explain how to train perceptron in detail. [9]

OR

- Q4)** a) What are two paradigms for Parallel Processing of neural network. State the scenario when to use which type of paradigm. [9]
b) Write a short note on : [9]
i) Multilayer perceptrons
ii) MLP as a Universal Approximator

P.T.O.

Q5) a) Using K-means clustering algorithm, cluster following data into two cluster. {2, 4, 10, 12, 3, 20, 30, 11, 15} Explain each step in detail. [9]

b) Write a short note on : [8]

i) Self - Organizing Maps

ii) PCA - Spectral Clustering

OR

Q6) a) A database has five transactions. min sup = 40% and confidence = 40%. [8]

TID Items Bought

T1 A, B, C

T2 A, B, C, D, E

T3 A, C, D

T4 A, C, D, E

T5 A, B, C, D

Find all frequent itemsets using Apriori algorithm.

b) Write and explain K-means clustering algorithm with example. [9]

Q7) a) Why is naive Bayesian classification called “naive”? Explain naive Bayesian classification algorithm in detail. [9]

b) What is HMM? Explain three Basic Problems of HMMs in detail. [9]

OR

Q8) a) What is regression and state its applications. Find linear regression equation for the following two sets of data : [9]

x	2	4	6	8
y	3	7	5	10

b) Write a short note on : [9]

i) d-Separation

ii) Junction Trees

